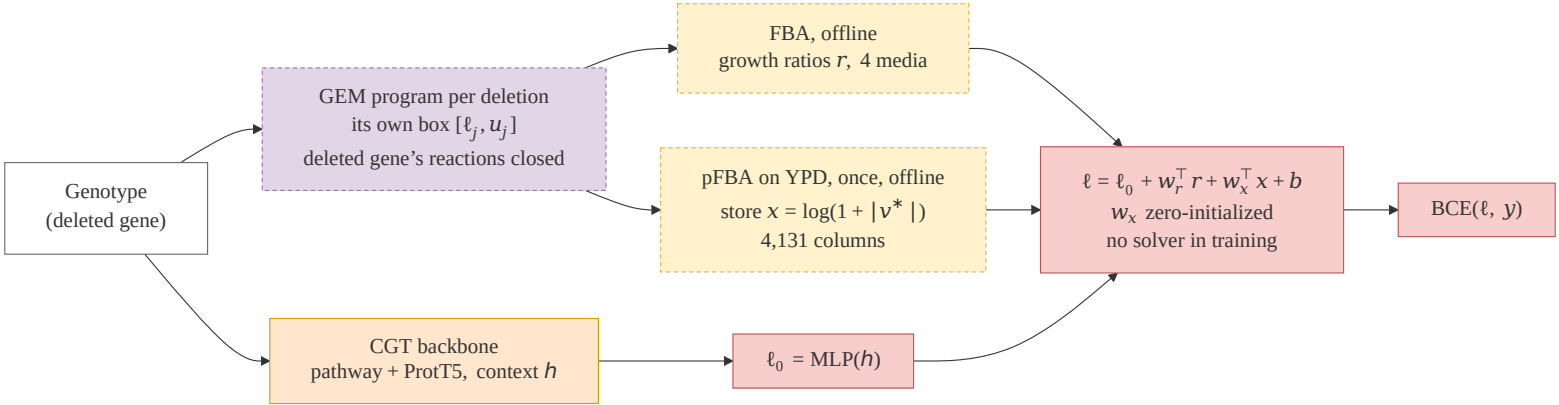

Genotype
(deleted gene)
GEM program per deletion
its own box $[\ell_j, u_j]$
deleted gene's reactions closed
FBA, offline
growth ratios r, 4 media
pFBA on YPD, once, offline
store $x = \log(1 + |v^*|)$
4,131 columns
CGT backbone
pathway + ProtT5, context h
ℓ_0 = MLP(h)
$\ell = \ell_0 + w_r^\top r + w_x^\top x + b$
w_x zero-initialized
no solver in training
BCE(ℓ, y)